

CALCULUS I 2019 Spring Midterm Exam	Dept. or School	Year	proctor
Student ID	Name		

* Your answer must be provided with descriptions how to get the answer.

1. (a) (2 points) Find the exact value of the following expression.

$$\cosh\left[\tanh^{-1}\left(\frac{1}{2}\right)\right] \times \cosh\left[\tanh^{-1}\left(\frac{1}{3}\right)\right] - \sinh\left[\tanh^{-1}\left(\frac{1}{2}\right)\right] \times \sinh\left[\tanh^{-1}\left(\frac{1}{3}\right)\right].$$

(풀이)

$$\cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y \text{ 이므로}$$

$$\therefore (\text{준식}) = \cosh\left[\tanh^{-1}\left(\frac{1}{2}\right) - \tanh^{-1}\left(\frac{1}{3}\right)\right]$$

$$\text{특히, } \tanh^{-1}x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \text{ 단, } |x| < 1,$$

$$\therefore \tanh^{-1}\left(\frac{1}{2}\right) = \ln\sqrt{3}, \quad \tanh^{-1}\left(\frac{1}{3}\right) = \ln\sqrt{2}$$

(1점)

$$\therefore (\text{준식}) = \cosh[\ln\sqrt{3} - \ln\sqrt{2}] = \cosh\left(\ln\frac{\sqrt{3}}{\sqrt{2}}\right)$$

$$= \frac{e^{\ln\frac{\sqrt{3}}{\sqrt{2}}} + e^{-\ln\frac{\sqrt{3}}{\sqrt{2}}}}{2} = \frac{\frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{2}}{\sqrt{3}}}{2}$$

$$= \frac{5}{2\sqrt{6}}. \quad (1\text{점})$$

(b) (3 points) If $f(x) = \sqrt{\frac{1+\tanh x}{1-\tanh x}} - \sinh\left(\frac{x}{4}\right) \cosh\left(\frac{x}{4}\right)$, find $f'(\ln 9)$.

(풀이)

$$1) \sqrt{\frac{1+\tanh x}{1-\tanh x}} = \sqrt{\frac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}}} = \sqrt{\frac{\cosh x + \sinh x}{\cosh x - \sinh x}} = \sqrt{\frac{e^x}{e^{-x}}} = e^{\frac{1}{2}x},$$

$$\therefore \frac{d}{dx} \sqrt{\frac{1+\tanh x}{1-\tanh x}} = \frac{d}{dx} \left(e^{\frac{x}{2}}\right) = \frac{1}{2} e^{\frac{x}{2}}, \quad (1\text{점})$$

$$2) \frac{d}{dx} \sinh\left(\frac{x}{4}\right) \cosh\left(\frac{x}{4}\right) = \frac{d}{dx} \frac{1}{2} \sinh\left(2 \times \frac{x}{4}\right) = \frac{1}{4} \cosh\left(\frac{x}{2}\right) \quad (1\text{점})$$

$$(\because \sinh(2\alpha) = 2 \sinh \alpha \cosh \alpha),$$

$$\therefore \frac{d}{dx} \sinh\left(\frac{x}{4}\right) \cosh\left(\frac{x}{4}\right) \Big|_{x=\ln 9} = \frac{1}{4} \cosh\left(\frac{\ln 9}{2}\right) = \frac{1}{4} \left(\frac{e^{\ln 3} + e^{-\ln 3}}{2}\right)$$

$$1), 2) \text{으로부터, } \therefore f'(\ln 9) = \frac{1}{2} e^{\frac{1}{2} \ln 9} - \frac{1}{4} \left(\frac{3+\frac{1}{3}}{2}\right) = \frac{3}{2} - \frac{5}{12} = \frac{13}{12}. \quad (1\text{점})$$

2. (3 points) (a) Find $\lim_{x \rightarrow \infty} (\ln x)^{\frac{\ln x}{x}}$.

(풀이)

$$\lim_{x \rightarrow \infty} (\ln x)^{\frac{\ln x}{x}} = [\infty]^0 \quad \because \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \text{ by } L'H.$$

$$\lim_{x \rightarrow \infty} (\ln x)^{\frac{\ln x}{x}} = \lim_{x \rightarrow \infty} e^{\ln(\ln x) \cdot \frac{\ln x}{x}} = \lim_{x \rightarrow \infty} e^{\frac{\ln x \times \ln(\ln x)}{x}} = e^{\lim_{x \rightarrow \infty} \frac{\ln x \times \ln(\ln x)}{x}}$$

$$\left(\because \lim_{x \rightarrow \infty} \frac{\ln x \times \ln(\ln x)}{x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{\left(\frac{1}{x}\right) \ln(\ln x) + \ln x \left(\frac{1}{\ln x}\right) \left(\frac{1}{x}\right)}{1}, \right.$$

$$= \lim_{x \rightarrow \infty} \frac{\ln(\ln x) + 1}{x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{\left(\frac{1}{\ln x}\right) \left(\frac{1}{x}\right)}{1} = 0.$$

$$\therefore (\text{준식}) = e^0 = 1. \quad (\text{단순 계산실수 } -1\text{점})$$

(b) (2 points) Find linearization $L(x)$ of $f(x) = \sec x$ at $a=0$ and use it to estimate $\sec\left(\frac{1}{\sqrt{2019}}\right)$.

(풀이)

$$f(x) = \sec x, \quad \frac{1}{\sqrt{2019}} \approx 0, \quad L(x) = f(0) + f'(0)(x-0).$$

$$\text{We have } f(0) = 1, \quad f'(x) = \sec x \tan x \text{ and so } f'(0) = 0$$

$$\text{and finally } L(x) = 0(x-0) + 1 = 1 \quad (1\text{ 점}). \text{ Now}$$

$$\therefore \sec\left(\frac{1}{\sqrt{2019}}\right) \approx L\left(\frac{1}{\sqrt{2019}}\right) = 1. \quad (1\text{점})$$

3. (4 points) Find $\lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt$.

(풀이)

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt &= \lim_{x \rightarrow 0^+} \frac{\int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt}{\sqrt{x^3}} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sqrt{x}} \sin(4x) - \frac{1}{2\sqrt{x}} \sin x}{\frac{3}{2} \sqrt{x}} \quad (\text{by L'H, FTC}) \\ &\dots\dots\dots(2\text{점}) \\ &= \frac{1}{3} \lim_{x \rightarrow 0^+} \frac{2\sin(4x) - \sin x}{x} \\ &= \frac{1}{3} \lim_{x \rightarrow 0^+} (8\cos(4x) - \cos x) \\ &= \frac{7}{3}. \dots\dots\dots(2\text{점}) \quad (\text{단순 계산실수 -1점}) \end{aligned}$$

4. (4 points) The solution between 3 and 4 of the equation $\tan x = 0$ is π . Starting with $x_1 = 3$, find the second approximation x_2 to π . (Use $\sin 6 = -0.30$)

(풀이)

Let $f(x) = \tan x$. Then $f'(x) = \sec^2 x$.

$$\begin{aligned} \therefore x_{n+1} &= x_n - \frac{\tan(x_n)}{\sec^2(x_n)} \\ &= x_n - \frac{\frac{\sin(x_n)}{\cos(x_n)}}{\frac{1}{\cos^2(x_n)}} \\ &= x_n - \sin(x_n) \cos(x_n) \\ &= x_n - \frac{1}{2} \sin(2x_n). \end{aligned}$$

$\dots\dots\dots(2\text{점})$

Since $x_1 = 3$,

$$\therefore x_2 = 3 - \frac{1}{2} \sin 6 = 3 - \frac{1}{2}(-0.30) = 3.15.$$

$\dots\dots\dots(2\text{점}) \quad (\text{단순 계산실수 -1점})$

5. Let $I_n = \int_0^1 (1-x^2)^n dx$ where n is a positive integer.

(a) (2 points) Find the number k such that $I_{20} = kI_{19}$.

(풀이)

$$\begin{aligned} I_n &= \left[x(1-x^2)^n \right]_0^1 + 2n \int_0^1 (1-x^2)^{n-1} x^2 dx \\ &= 2n(I_{n-1} - I_n) \\ \therefore I_n &= \frac{2n}{2n+1} I_{n-1}, \quad \therefore k = \frac{40}{41}, \quad \dots\dots\dots(2\text{점}) \end{aligned}$$

(b) (3 points) Find I_{20} .

(풀이)

$$\therefore I_{20} = \frac{2^{40}(20!)^2}{41!} \quad \dots\dots\dots(3\text{점}) \quad \underline{\text{(단순 계산실수 -1점)}}$$

6. (4 points) Suppose the graph of the function $y = x^3 + px^2 + qx$ intersects the parabola $y = x^2$ when $x=0$, $x=1$ and $x=a$, where $a > 1$. If the two regions between the curves have the same area, find the volume of the solid obtained by rotating the two regions about the y -axis.

(풀이)

$$\text{Since } \int_0^a (x^3 + px^2 + qx - x^2) dx = 0, \quad a=2, p=-2, q=2. \quad \dots\dots\dots(1\text{점})$$

The volume

$$\begin{aligned} \therefore V &= \int_0^1 2\pi x [(x^3 - 2x^2 + 2x) - x^2] dx + \int_1^2 2\pi x [x^2 - (x^3 - 2x^2 + 2x)] dx \\ &= 2\pi \int_0^1 x(x^3 - 3x^2 + 2x) dx + 2\pi \int_1^2 x(-x^3 + 3x^2 - 2x) dx \\ &= 2\pi \left(\frac{1}{5} - \frac{3}{4} + \frac{2}{3} \right) + 2\pi \left[-\frac{1}{5}(2^5 - 1) + \frac{3}{4}(2^4 - 1) - \frac{2}{3}(2^3 - 1) \right] \\ &= \pi. \quad \dots\dots\dots(2\text{점}) \quad \underline{\text{(단순 계산실수 -1점)}} \end{aligned}$$

7. (3 points) Evaluate the indefinite integral:

$$\int \frac{1}{x\sqrt{3x^2 - 2x - 1}} dx. \quad (\text{Hint: Substitute } x = \frac{1}{t})$$

(풀이)

$$\text{Let } x = \frac{1}{t}. \text{ Then } dx = -\frac{1}{t^2} dt$$

$$\therefore (\text{준식}) = \int \frac{1}{\left(\frac{1}{t}\right)\sqrt{3\left(\frac{1}{t}\right)^2 - 2\left(\frac{1}{t}\right) - 1}} \left(-\frac{1}{t^2}\right) dt = -\int \frac{1}{\sqrt{3-2t-t^2}} dt$$

$$\text{여기서, } \sqrt{3-2t-t^2} = \sqrt{4-(t+1)^2}, \quad t+1 = 2\sin\theta$$

$$\therefore (\text{준식}) = -\int \frac{1}{2\cos\theta} (2\cos\theta d\theta) = -\int d\theta = -\theta + C \quad \dots\dots\dots(2\text{점})$$

$$\begin{aligned} \text{즉, } \int \frac{1}{x\sqrt{3x^2 - 2x - 1}} dx &= -\sin^{-1} \left[\frac{1}{2} \left(\frac{1}{x} + 1 \right) \right] + C \\ &= -\sin^{-1} \left(\frac{1+x}{2x} \right) + C. \quad \dots\dots\dots(1\text{점}) \end{aligned}$$

8. (5 points) Evaluate the definite integral:

$$\int_0^{\pi} \left[\frac{\cosh x}{\cosh x + \cosh(\pi-x)} + \frac{x \sin x}{1 + \cos^2 x} \right] dx.$$

(풀이)

(1) Let $x = \pi - u$. Then $dx = -du$

$$\begin{aligned} I &= \int_0^{\pi} \left[\frac{\cosh x}{\cosh x + \cosh(\pi-x)} \right] dx = \int_{\pi}^0 \left[\frac{\cosh(\pi-u)}{\cosh(\pi-u) + \cosh u} \right] (-du) \\ &= \int_0^{\pi} \left[\frac{\cosh(\pi-u)}{\cosh(\pi-u) + \cosh u} \right] du \end{aligned}$$

$$\therefore 2I = \int_0^{\pi} \left[\frac{\cosh x + \cosh(\pi-x)}{\cosh x + \cosh(\pi-x)} \right] dx = \int_0^{\pi} 1 dx = \pi, \quad \therefore I = \frac{\pi}{2}.$$

.....(2점)

(2)

$$\begin{aligned} \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \int_{\pi}^0 \frac{(\pi-u) \sin(\pi-u)}{1 + \cos^2(\pi-u)} (-du) \\ &= \int_0^{\pi} \frac{(\pi-u) \sin(\pi-u)}{1 + \cos^2(\pi-u)} du = \int_0^{\pi} \frac{(\pi-u) \sin(u)}{1 + \cos^2(u)} du \\ &= \int_0^{\pi} \frac{\pi \sin(u)}{1 + \cos^2(u)} du - \int_0^{\pi} \frac{u \sin(u)}{1 + \cos^2(u)} du \end{aligned}$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

.....(1점)

let $u = \cos x$ then $du = -\sin x dx$

$$\begin{aligned} \therefore (2) &= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_1^{-1} \frac{du}{1+u^2} (-du) \\ &= \frac{\pi}{2} \int_{-1}^1 \frac{du}{1+u^2} (du) = \frac{\pi}{2} [\tan^{-1} u]_{-1}^1 = \frac{\pi^2}{4}. \end{aligned}$$

.....(2점)

$$\therefore \int_0^{\pi} \left[\frac{\cosh x}{\cosh x + \cosh(\pi-x)} + \frac{x \sin x}{1 + \cos^2 x} \right] dx = \frac{\pi}{2} + \frac{\pi^2}{4}.$$

9. (5 points) Evaluate the indefinite integral:

$$\int \left(\frac{1}{\sin^2 x \cos^2 x} + \frac{\sec^2 x}{\tan^3 x - 1} \right) dx.$$

(풀이)

$$\begin{aligned} (1) \int \frac{1}{\sin^2 x \cos^2 x} d\theta &= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} dx \\ &= \int (\sec^2 x + \csc^2 x) dx = \tan x - \cot x + C_1 \end{aligned}$$

.....(2점)

$$\begin{aligned} (2) \int \frac{\sec^2 x}{\tan^3 x - 1} dx &= \int \frac{1}{t^3 - 1} dt \quad (\because \tan x = t) \\ \frac{1}{t^3 - 1} &= \frac{1}{(t-1)(t^2+t+1)} = \frac{1}{3} \left(\frac{1}{t-1} - \frac{t+2}{t^2+t+1} \right) \\ &= \frac{1}{3} \left(\frac{1}{t-1} \right) - \frac{1}{6} \left(\frac{2t+1}{t^2+t+1} \right) - \frac{3}{6} \left(\frac{1}{t^2+t+1} \right) \end{aligned}$$

$$\therefore (2) = \int \frac{1}{3} \left(\frac{1}{t-1} \right) dt - \int \frac{1}{6} \left(\frac{2t+1}{t^2+t+1} \right) dt - \int \frac{1}{2} \left(\frac{1}{t^2+t+1} \right) dt$$

.....(1점)

$$\therefore (2) = \int \frac{1}{3} \left(\frac{1}{t-1} \right) dt - \int \frac{1}{6} \left(\frac{2t+1}{t^2+t+1} \right) dt - \int \frac{1}{2} \left(\frac{1}{t^2+t+1} \right) dt$$

$$= \frac{1}{3} \ln |t-1| - \frac{1}{6} \ln |t^2+t+1| - \frac{1}{2} \int \frac{1}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} dt$$

$$\text{여기서, } t + \frac{1}{2} = \frac{\sqrt{3}}{2} \tan \theta, \quad dt = \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$\therefore -\frac{1}{2} \int \frac{1}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} dt = -\frac{1}{2} \left(\frac{2}{\sqrt{3}} \right) \int \frac{1}{\tan^2 \theta + 1} (\sec^2 \theta d\theta)$$

$$= -\frac{1}{\sqrt{3}} \int d\theta = -\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{3}} \right) + C_2$$

.....(2점)

(1), (2),

$$\therefore (\text{준식}) = (\tan x - \cot x) +$$

$$\left[\frac{1}{3} \ln |\tan x - 1| - \frac{1}{6} \ln |\tan^2 x + \tan x + 1| - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{3}} \right) \right] + C$$

(1) 별해:

$$\int \frac{1}{\sin^2 x \cos^2 x} d\theta = \int \frac{1}{\left(\frac{1}{2} \sin 2x\right)^2} dx = 4 \int \frac{1}{1 - \cos 4x} dx$$

$$= 2 \int \frac{1}{1 - \cos u} du \quad (\because 4x = u)$$

$$= 2 \int \frac{1}{1 - \left(\frac{1-t^2}{1+t^2}\right)} \left(\frac{2}{1+t^2} dt \right) \quad (\because \tan \left(\frac{u}{2}\right) = t)$$

$$= 2 \int \frac{1}{t^2} dt = -2 \frac{1}{\tan \left(\frac{4x}{2}\right)} + C = -2 \cot(2x) + C$$